Application of Stack

Expressions evaluation

Expression Evaluation

- The priority of the operations

1. •Parenthesis
2. •The power
3. •Multiply / division
4. •Adding / subtraction

Example

3 * 5/ (52*4-3) +10 / 2

Sorting Operands and operations (3+5)

Infix 3 + 5

- **Postfix** 3 5 +
- **Prefix** + 3 5

I don't understand anything

Example 1

Infix 4 + 7 * 2

Postfix (4) + (7*2)

(4) +(7 2 *)

(4) (7 2 *) +

the final result is

4 7 2 * +

14

Arrange items
Postfix

4+14 = 18

Example 1

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Postfix (4) + (7*2)

(4) +(7 2 *)

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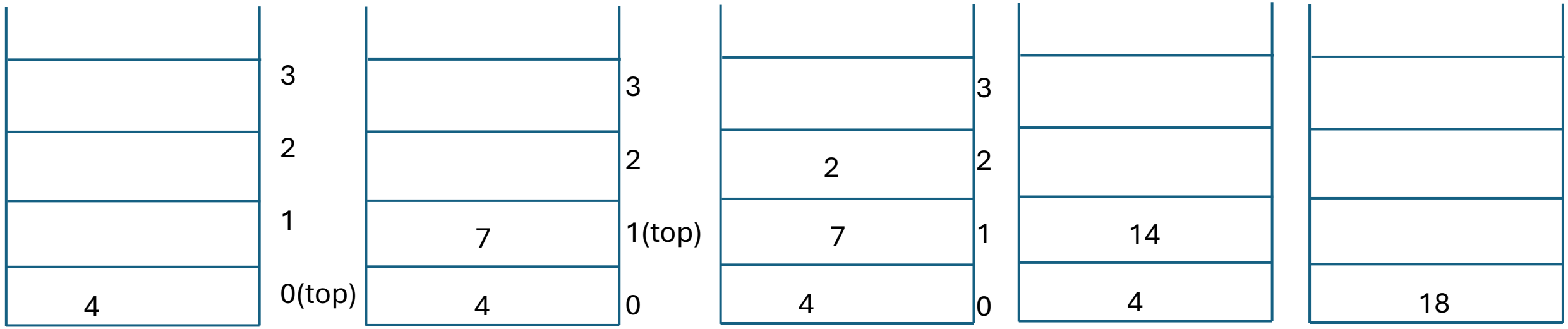


Example 2

- Infix: A/B + 4
- Postfix: (A/B) + (4)

  (A B /) + (4)

  (A B /) (4) +

The final result : A B / 4 +

Example 3

- Infix: (5+2) * 3
- Postfix:

(5+2) * (3)

(5 2 +) (3) *

The final result : 5 2 + 3 *

Example 4

- Infix : (A*B) + (C/ D)
- Postfix:

(A*B) + (C/D)

(A B *) + (C D /)

(A B*) (C D /) +

The final result : A B * C D / +

Example 5

- Infix: 5/5+2-1*9
- Postfix:

(5/5)+(2)-(1*9)

(5 5 /) + (2) – (1 9 *)

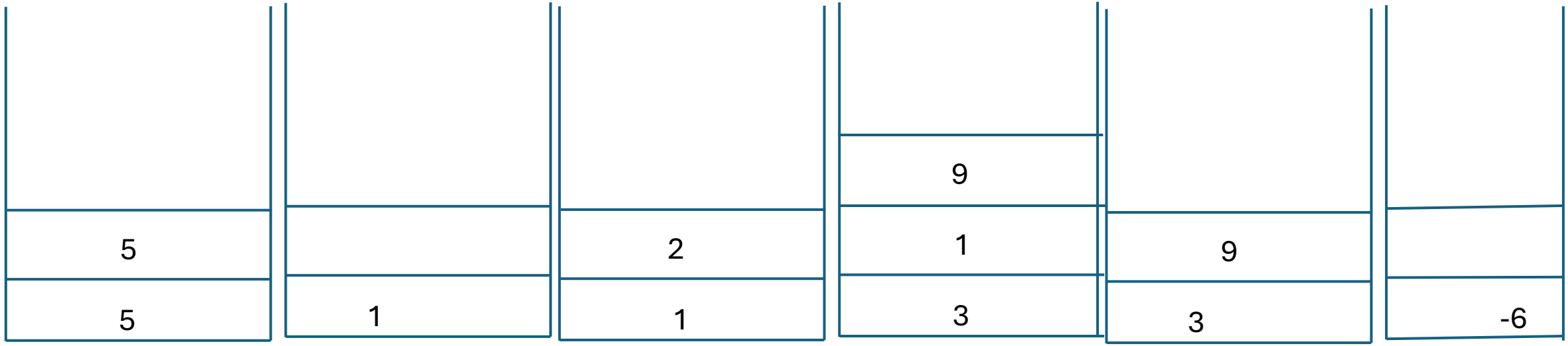
(5 5 /) (2) + (1 9 *) –

the final result

5 5 / 2 + 1 9 * -

How to use stack in the postfix
(5 5 / 2 + 1 9 * -)

- The stack work from lest to write → (5 5 / 2 + 1 9 * -)



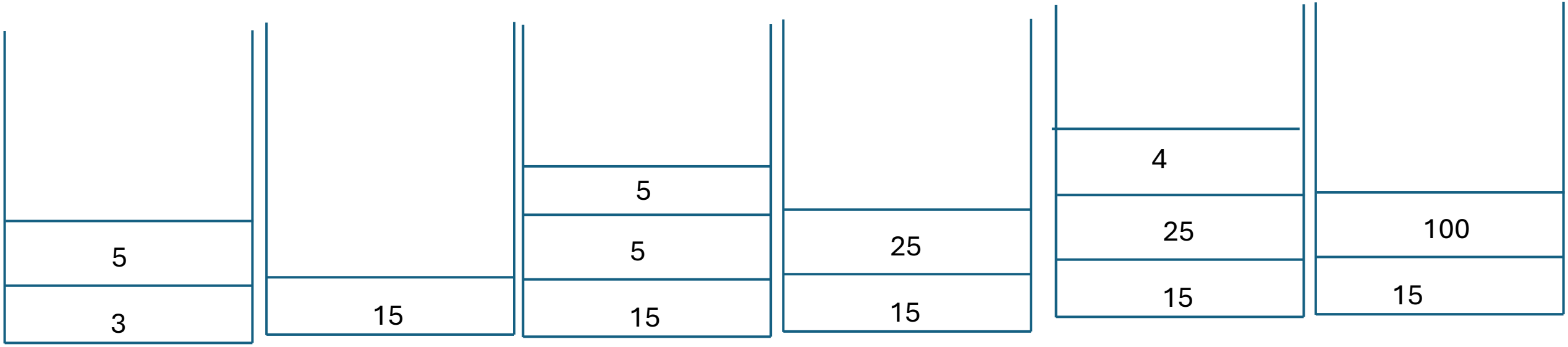
Example:

$$3 * 5/ (5^2*4-3) +10 / 2$$

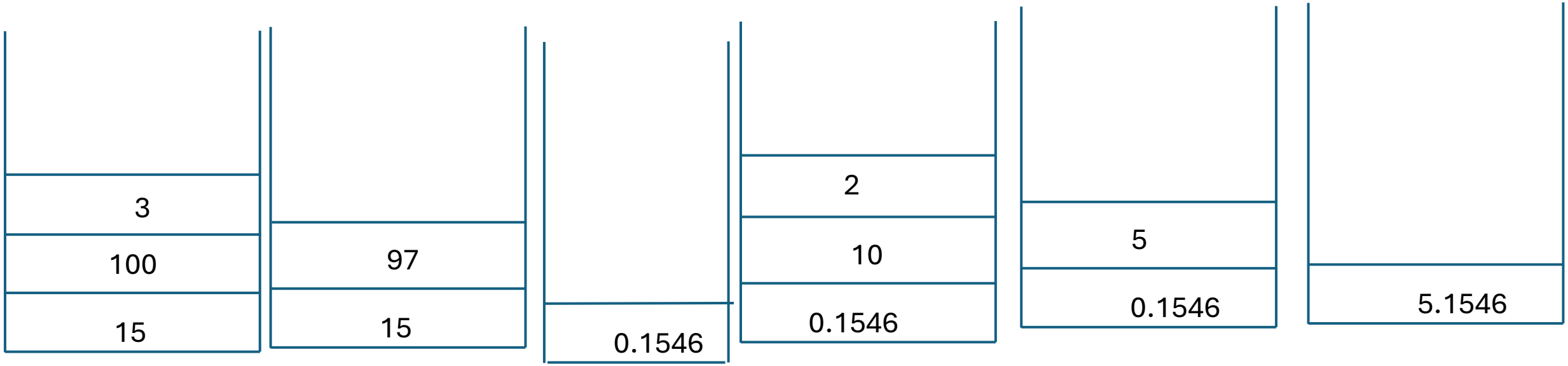
Example:

$$3 * 5/ (5^2*4-3) +10 / 2$$

- $(3*5)/ \ (((5^2)*4) – 3) + (10/2)$
- (3 5 *) /(((5*5) * 4)-3)+(10/2)
- (3 5 *) / (((5 5 *) *4) -3)+(10 2 /)
- (3 5 *) / ((5 5 * 4 *) -3) + (10 2 /)
- (3 5 *) / (5 5 * 4 * 3 -) + (10 2 /)
- 3 5 * 5 5 * 4 * 3 - / 10 2 / +


5
3
15
5
5
15
25
15
4
25
15
100
15

3 5 * 5 5 * 4 * 3 - / 10 2 / +


3
100
15
97
15
0.1546
2
10
0.1546
5
0.1546
5.1546

Postfix method

- Example: 4-6*5/8+2
- Solution: (4)- ((6*5)/(8)) +(2)
- = (4) – ((65*)/(8))+(2)
- =(4)-(65*8/)+(2)
- =465*8/-2+

- Start from left to the RIGHT in postfix.
- The less index is the first and high index is the second in subtraction and division.

5		8				
6	30	30	30/8=3.75		2	
4	4	4	4	0.25	0.25	2.25

The power in postfix

- Example:

8*5+3^2-4

= (8*5) + (3^2) – (4)

= (85*)+(32^)- (4)

= 85*32^+4-

		2				
5		3	9		4	
8	40	40	40	49	49	45

Prefix method

- 4-6*5/8+2

= (4)-((6*5)/8)+(2)

=(4)-((*65)/8)+(2)

=(4)-(/*658)+(2)

= +-4/*6582

- Start from **RIGHT** to the **LEFT** in postfix.
- The **HIGH** index is the **first** and **LESS** index is the **second** in subtraction and division.

6						
5	30			4		
8	8	30/8=3.75	3.75	3.75	0.25	
2	2	2	2	2	2	2.25

The power in prefix

- Example:

8*5+3^2-4

= (8*5) + (3^2) – (4)

= (*85)+(^32)- (4)

= -+*85^324

		8			
3		5	40		
2	9	9	9	49	
4	4	4	4	4	45